Worksheet 6.1 — Introduction to Vectors

Grade 12 Calculus and Vectors (MCV4U) · Unit 6: Geometry and Algebra of Vectors

The Big Ideas

A vector has magnitude and direction (velocity, force); a scalar has magnitude only (speed, mass). Equal vectors share magnitude and direction; $-\vec{v}$ reverses direction.

- Vector = magnitude + direction; scalar = magnitude only.
- Equal vectors: same magnitude and direction (position-free).
- Directions via bearings ($N30°E$, 030°).

The Big Picture

Magnitude and direction

A **vector** has both size and direction, unlike a scalar. Displacement, velocity, and force are vector quantities.

Representing vectors

Draw a vector as a directed arrow; two vectors are equal when they share magnitude and direction, wherever they are placed.

Special vectors

The **zero vector** has no length, and the opposite of a vector reverses its direction.

Worked Examples

Example 1 — Vector or scalar

Classify velocity, speed, force, temperature, displacement.

Solution. **Step 1 (ask whether direction matters for each quantity):** velocity, force, and displacement all depend on which way they point; speed and temperature do not.

Conclusion: vectors: velocity, force, displacement. Scalars: speed, temperature.

Example 2 — Magnitude & direction

A plane flies 500 km at $N60°E$.

Solution. **Step 1 (identify the magnitude):** the distance travelled is 500 km.

Step 2 (identify the direction): the bearing $N60°E$ means 60° measured from north, toward east.

Conclusion: magnitude 500 km; direction $N60°E$.

Example 3 — Equal vectors

When are two displacement vectors equal?

Solution. **Step 1 (recall what defines a vector):** magnitude and direction, not position.

Conclusion: two vectors are equal exactly when they share magnitude and direction — where they're drawn doesn't matter.

Example 4 — Opposite vector

If $\vec{v}$ is 8 m/s east, describe $-\vec{v}$.

Solution. **Step 1 (recall what $-\vec{v}$ means):** same magnitude, reversed direction.

Step 2 (apply it): magnitude stays 8 m/s; east reverses to west.

Conclusion: $-\vec{v}$ is 8 m/s west.

Example 5 — Bearing

Express "40° east of north" as a true bearing.

Solution. **Step 1 (recall true bearing format):** measured clockwise from north, as a 3-digit angle.

Step 2 (convert): "40° east of north" is already 40° clockwise from north.

Conclusion: 040°.

Example 6 — Vector or scalar

Is acceleration a vector or scalar?

Solution. **Step 1 (ask if direction matters):** acceleration has a direction (the way the velocity is changing).

Conclusion: vector.

Example 7 — Vector or scalar

Is mass a vector or scalar?

Solution. **Step 1 (ask if direction matters):** mass is just an amount, with no direction.

Conclusion: scalar.

Example 8 — Opposite

Describe $-\vec{v}$ if $\vec{v}$ is 10 N north.

Solution. **Step 1 (keep the magnitude, reverse the direction):** 10 N stays the same; north reverses to south.

Conclusion: $-\vec{v}$ is 10 N south.

Example 9 — Bearing

Write $N75°E$ as a true bearing.

Solution. **Step 1 (convert from N-E form to true bearing):** $N75°E$ means 75° clockwise from north.

Conclusion: 075°.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Is force a vector or a scalar?

Question 2

Is temperature a vector or a scalar?

Question 3

Describe $-\vec{v}$ if $\vec{v}$ is 6 m/s west.

Question 4

Two vectors have equal magnitude, opposite direction. Equal?

Question 5

Write $N20°E$ as a true bearing.

Question 6

Is displacement a vector or a scalar?

Question 7

Is speed a vector or a scalar?

Question 8

Describe $-\vec{v}$ if $\vec{v}$ is 12 N east.

Question 9

What two things define a vector?

Question 10

Write $N50°E$ as a true bearing.

Question 11

Is energy a vector or a scalar?

Question 12

Does the position of a vector affect whether two are equal?

Question 13 — Challenge

A ship sails 30 km at a bearing of 120°. State its magnitude and describe the bearing in NE form.

Question Recap & Answer Key

Question Recap

1. Is force a vector or a scalar?
2. Is temperature a vector or a scalar?
3. Describe $-\vec{v}$ if $\vec{v}$ is 6 m/s west.
4. Two vectors have equal magnitude, opposite direction. Equal?
5. Write $N20°E$ as a true bearing.
6. Is displacement a vector or a scalar?
7. Is speed a vector or a scalar?
8. Describe $-\vec{v}$ if $\vec{v}$ is 12 N east.
9. What two things define a vector?
10. Write $N50°E$ as a true bearing.
11. Is energy a vector or a scalar?
12. Does the position of a vector affect whether two are equal?
13. A ship sails 30 km at a bearing of 120°. State its magnitude and describe the bearing in NE form.

✓ Answer Key

1. vector
2. scalar
3. 6 m/s east
4. no (opposites)
5. 020°
6. vector
7. scalar
8. 12 N west
9. magnitude and direction
10. 050°
11. scalar
12. no
13. 30 km; $120° = S60°E$